Perspective and orthogonal CBCT/CT digitally reconstructed radiographs compared to conventional cephalograms

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***Abstract*—The aim of this paper is to compare conventional radiographs, with those synthesized from CBCT/CT volumes. Four methods for generating Digitally Reconstructed Radiographs (*DRR's*) were compared: Ray-Sum, Radon Transform, Siddon's Algorithm and an hybrid approach. Three random skulls were used as the CT/CBCT input, each one with their conventional cephalometric lateral radiograph to validate results. Volume data in *DICOM* format were imported into Matlab, and orthogonal and perspective lateral projection were created. The PSNR (peak signal-to-noise ratio) measured the difference between DRR's and conventional digital radiographs. Finally, applications of the most similar *DRR* to a conventional cephalogram are discussed, considering that the use of CBCT images in orthodontics also allows 3D cephalometry and provides full information about the structure of each patient.**

***Index Terms*—reconstructed radiography, cone beam tomography, cephalometry, projections**

I. Introduction

A cephalogram is primarily used to describe the morphology of the craniofacial skeleton in orthodontics. A cephalometry is a procedure in which human skull measures are defined from lateral cephalogram [16], is a standardized method that is part of the diagnostic, treatment and planning process in different types of maxillofacial surgery and orthodontics. Recently, Computed Tomography (CT) and Cone Beam CT (CBCT), were introduced to the dental community as a diagnostic tool and has became a standard imaging technique in orthodontics [10], [16]. This is because tomography scans provide accurate 3D information of size and position of the patient volume. Both, CT and CBCT images allow reforming the three-dimensional structure of the skull into 2D conventional simulated X-ray images or *DRR's* for cephalometry, keeping size and position, then *DRR's* can be defined as a 2-D simulated approximation of a radiograph [2]. *DRR* rendering, is a DVR (Direct Volume Rendering) technique consisting in virtual x-rays passing through a reconstructed CT volume and is also called simulated X-ray volume rendering.

Nowadays, three-dimensional arrays of data are usually generated by Cone Beam Computed Tomography (CBCT) to be analyzed and visualized by volume rendering techniques to ease interpretation. These methods allow experts to see interior structures and spatial relationships of a patient skull. Then, cephalometry can be performed by using not only a single X-ray image, e.g. Moshiri *et al.* [9] presented a cephalometry method for *DRR's* with CBCT showing the importance of using *DRR* for this procedure. Figure 1.1 shows the general scheme for DVR by using the original slices of a CT scan viewed as a three-dimensional array that can be processed to obtain a *DRR*. Finally, the main problem of this ray-driven methods is still connected with a long computation time especially in CBCT high resolution sets [8].

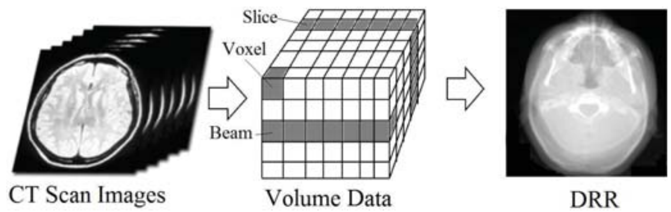

Figure 1.1. Direct volume rendering scheme. From volumetric CT scan formed by a set of N images a 3D array of voxels can be created and finally by using a ray casting algorithm for rendering, a new image of a reconstructed X-ray image is created.

II. Materials and Methods

In this study, we used the examinations of the head of three patients, two head CBCT volumes provided by the Faculty of Odontology, Autonomous University of the State of Mexico and a CT public volume dataset provided in the DICOM sample image sets from OsiriX [11] (MANIX). The CBCT volumes consists of 528 and 503 slices and isometric voxel 0.4mm size. The third volume is a CT scan consisting of non-isometric voxels with 460 slices separated for 0.7 mm. All volumes are in DICOM format and were used and analyzed to test the reconstruction algorithms. Figure 2.1 shows the Multiplanar Reconstructions (MPR) of the used volumes, rendered in 3DSlicer. The DICOM data were loaded into Matlab without any preprocessing to generate *DRR's*. All volume keeps voxels with density values of the digitized material as it is. Then,, generation of *DRR* is treated as volume rendering or X-ray simulation. The technique consists of simulated x-rays passing through the MPR volume and is directly based on the law of attenuation absorption [7]. *DRR's* can be directly rendered from the volume by a variety of algorithms using the operations called *ray-driven*, as *DRR* rendering is a special case of volume rendering. Techniques by *ray-driven* for rendering *DRR* images can be grouped into two categories: image-based (backward-projective) or object-based (forward-projective) [14]. A 3D virtual patient head was created from each study and the Frankfort plane of each volume was oriented horizontally based on the sagittal plane. An orthogonal and perspective radiographs were built from the reoriented volumes. Parallel beam rays created the orthogonal projections and for perspective projections, the source of the rays was the center of projection (focus) 1000-1500 mm away from the projection plane. In perspective, the location of an object between the focus and the projection plane determines its size. A scheme is in Figure 2.5. The orthogonal radiographs were created with 0% magnification and perspective radiographs were created using 5-8% magnification in the mid-sagittal plane.

Figure 2.1. Volumes created from CBCT (left/center), and volume created from a conventional CT from the public dataset MANIX provided in the DICOM sample image sets from OsiriX [11] , all MPR in 3DSlicer [12].

A. Ray-sum Algorithm

Ray-Sum is a technique whereby hypothetical X-ray are sent from each pixel of a source towards the volume to a final image in the screen. The objective is to sum all the ray lengths through all the voxels in CT volume data multiplied by voxel densities to get the radiological path.

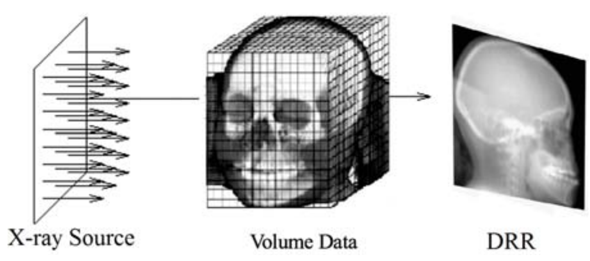

Figure 2.2. In Ray-Sum, 3D data (voxels) is displayed by the average of all intensities from the rays source to the projection plane forming the *DRR*.

The values associated with the voxels determine what happens to each ray and therefore what image is finally reconstructed. For each pixel of the final image on the screen, a ray is used to intersect parallel voxels in the volume (just one direction). Each pixel of the projection gets a 12-bit value by averaging all intensities of the intersected pixels. This view has a translucent appearance analogous to conventional radiographs. Figure 2.2 shows the scheme of the source of simulated rays, volume and detector configuration to get a *DRR*.

Ray-sum algorithm is a forward projection $\rho(i,j,k)$ denote the voxel attenuation (gray level intensity) in a 3-dimensional CT volume and $l(i,j,k)$ the length of the intersection of an X-ray with that voxel, then the radiological path is defined as:

$$d = \sum_i \sum_j \sum_k l(i,j,k)\rho(i,j,k) \tag{1}$$

The value d represents the sum of the contribution of the intensities associated to a density value (in Hounsfield units) of the simulated radiological path of the ray [13]. The simulated radiological path is parallel, and this approximation is away from the physics involved when an X-ray image is generated, but the resultant *DRR* has no distortion or magnification as in the conventional X-ray capture. Computing *DRR's* using this algorithm is $O(n^3)$ and may result inefficient because many voxels and paths values will be zero [1]. The *DRR* created by this projection must be post-processed with filters to get better contrast and sharpness in radio-lucid and radio-opaque zones to correctly detect cephalometric landmarks by an orhodoncist, surgeon or a computer program.

B. Radon transform

In this method, *DRR's* are generated from the CT volume data by computing the attenuation of a monoenergetic beam due to different material in the human body (e.g., bone, soft tissue, water, etc.).

$$I = I_0 exp - \int_0^D \mu(x)dx \tag{2}$$

By equation (2), the Beer's Law [4], the Radon transform is applied to the intensity image formed by each slice of the volume data i a discrete way. The Radon transform is the projection of the image intensity along a radial line oriented at a specific angle. For *DRR's,* a projection of a 2D function $f(x,y)$ is a set of line integrals, and the radon transform computes the line integrals from a source along parallel paths, or beams. The beams are spaced by each pixel unit in the detector. To represent a *DRR*, parallel-beam projections of an image from the same angle for each slice in volume data are calculated. The following Figure shows a single projection at a specified rotation angle.

When an individual ray has passed through the volume data, its value is:

$$I = I_0 \sum_{i=1}^{n} e^{-\mu_i d_i} \tag{3}$$

where I_0 is the original ray value, i is the voxel through which the ray passes, μ_i is the linear attenuation coefficient of the material in voxel i and d_i is the segment between the entrance and output point of the ray in voxel i. The attenuation coefficient of the material for each voxel can be obtained by:

$$CTnumber = 1000*[(\mu_i - \mu_w)/\mu_w] \tag{4}$$

where μ_w is the linear attenuation coefficient of water for the average energy in the CT beam.

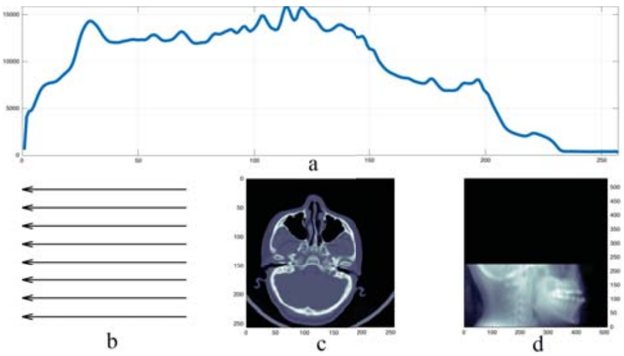

Figure 2.3. For each slice in the CT volume data, radon transform is used to obtain an attenuation profile (a) following the direction of the X-ray beam (b), in this case to reconstruct a sagittal projection. Image (c) is the actual slice and (d) is the set of the acumulated transforms that construct the *DRR*.

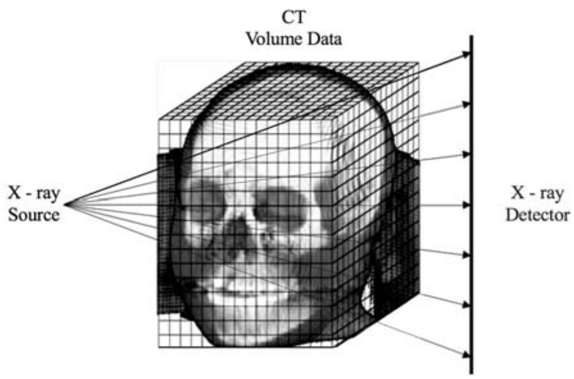

Figure 2.4. This scheme illustrates a 3D ray tracing though a volume data array. The lines from the simulated X-ray source on parametric planes indicate the tracing. The tracing can be done in any plane to get *DRR's* from any angle.

C. Siddon´s Algorithm

The equation (1) can be evaluated over all voxels in the volume, but results inefficient and time consuming, Siddon [15] proposed a more efficient method by viewing volume data voxels as the intersection of equally spaced, parallel planes. The intersection of the ray with the planes is then calculated, rather than the intersection of the ray with the different voxels. The intersection with the first plane is calculated and the rest follows at fixed intervals because the planes are equally spaced. The data in the CT array may be considered as the intersection areas of orthogonal sets of the equally spaced, parallel lines. Two equally spaced sets give the intersections of the ray with the lines: one set for the horizontal lines and another set for vertical lines [15].

D. Ray-Sum and Radon transform for cone beams.

The fourth method is an hybrid approach that takes algorithms A and B and transforms the parallel beam shape rays into a cone beam shape. A conventional radiograph capturing scene is reconstructed. Radiological paths are conducted from a source of X-rays centered on a point around 1000mm away from the volume centroid. A detector of nxn pixels is also placed from 300 to 500 mm of the centroid of volume in opposite side to the source. Figure 2.5 shows the scheme of

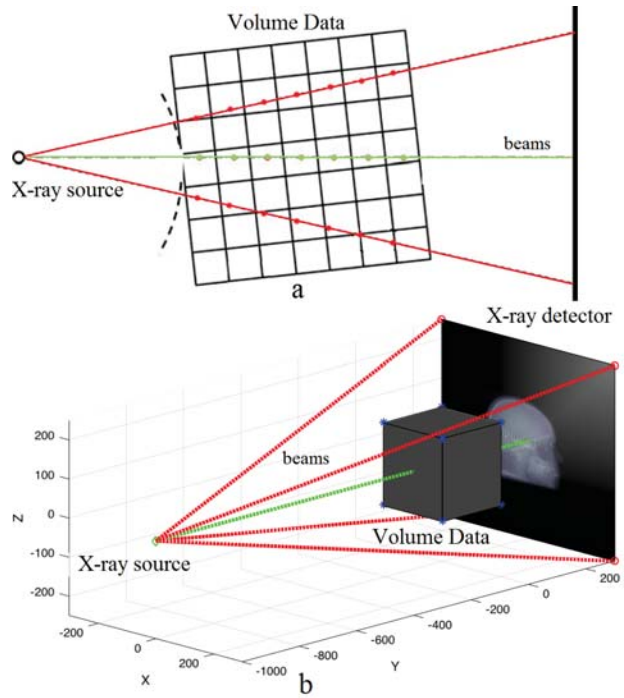

Figure 2.5. 2-D (a) and 3-D (b) illustrations of the configuration when X-rays in a cone beam intersect with a CT/CBCT volume. In some cases sample points can be located within the same voxel producing blurred projections.

the configuration used for generating *DRR* using this method. The magnification in *DRR's* was calculated using conventional digital radiographs parameters, the distance between the source and the mid-sagittal plane in the volume was 1500mm, and the distance between the detector and the mid-sagittal plane in the volume was 100mm [5]. Thus, the *DRR's* magnification was [5]:

$$Magnification = \frac{100}{1500}x100\% = 6.66\%$$

Perspective CBCT projections were adjusted for the 6.66% magnification to simulate conventional digital radiographs.

III. Results

The generation of a *DRR* comprises the calculation of line integrals over the Hounsfield values along the rays through the voxel volume. The rays are defined by the focal spot of the (virtual) X-ray source, and a discrete point on the (virtual) detector grid.

For the evaluation of the *DRR's*, we compute the difference between the outputs of intensity-based registrations of the calculated *DRR's* and the conventional X-ray image by applying a rigid transformation. Differences between radio-opaque and radio-lucid zones were the reference to determine which image could be better for cephalometry. Figure 3.4 shows this process, and the PSNR between the conventional digital Radiograph and all generated *DRR's*. PSNR and Mean Square Error (MSE) [3] were used to compare the squared error between the original image and the reconstructed image. There is an inverse relationship between PSNR and MSE. A higher PSNR value indicates the higher quality of the image. The answer in decibels (dB), in our tests using the method of the Radon

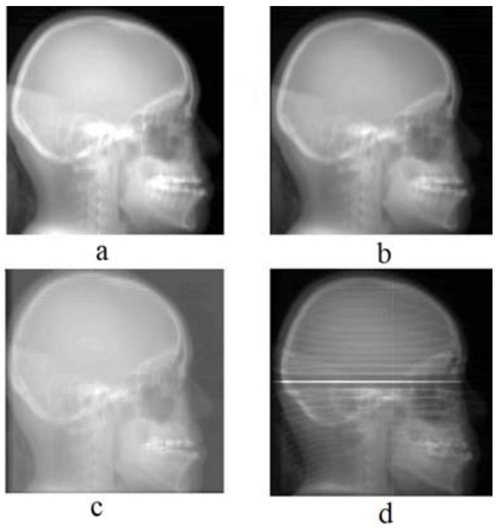

Figure 3.1. Example for *DRR* generation of a skull in its sagittal view from CT using the four methods (a) shows a sample using Ray-Sum, (b) shows the use of Radon transform, (c) shows the use of Siddon´s Algorithm and (d) the Radon transform with cone beams.

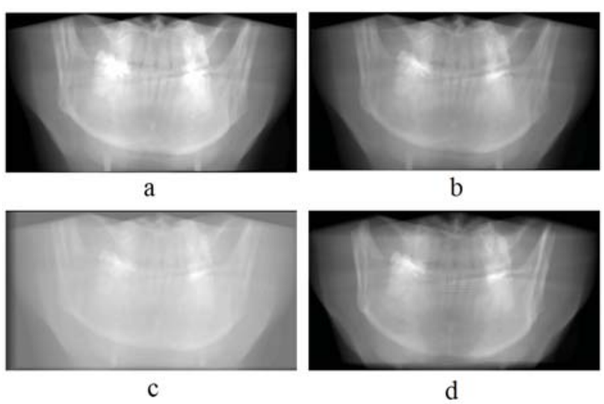

Figure 3.2. Example for *DRR* generation of mandible-maxilla in its coronal view from CBCT using the four methods (a) shows a sample using Ray-Sum, (b) shows the use of Radon transform, (c) shows the use of Siddon´s Algorithm and (d) the Radon transform with cone beams.

transform with perspective cone beams get the highest value with +10.86 db, that indicates this *DRR* is the most similar to the conventional digital radiograph. In the other hand, the next better value was +10.50 obtained by the method of the Radon transform with orthogonal parallel beams. It indicates that the generated *DRR* is very similar to the conventional digital radiograph but with the main difference that is has not magnification. The *orthogonal projections DRR's* shows that radiographs without magnification can be matched to its CT/CBCT volume and the cephalometric landmarks in the mid-sagittal plane should be the same, therefore, *orthogonal projections DRR's* could be used for the 2D conventional cephalometric analysis and in the future for 3D cephalometric analysis.

IV. Discussion

Before discussing our results, it is important to reiterate that the motivation to evaluate different methods for generating *DRR's* is to use them for cephalometry. In orthodontics lateral and frontal cephalograms, both with facial photographs are currently the main diagnostic imaging modalities for diagnostic. A CBCT/CT scan provides good quality medical information and also allows 3D cephalometry, providing to patients and practitioners a better understanding of the 3D skull structure. In conventional cephalograms, structures overlap by the projection onto a 2D plane. Using 3D CBCT images in cephalometry, has more advantages apart from the 2D reconstructed radiographs like reduced radiation exposure, natural shape of the soft-tissue facial mask, reduced occlusion and in-office imaging.

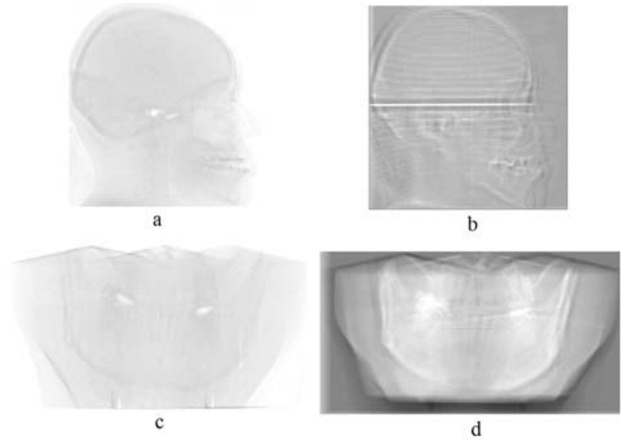

Figure 3.3. Image (a) shows the difference image, formed after registering images, by pixel subtraction between image 3.1a and 3.1b while (b) shows the difference image formed by pixel by pixel subtraction between image 3.1c and 3.1d. Image (c) shows the difference image, formed after registering images, by pixel by pixel subtraction between image 3.2a and 3.2b and (d) shows the difference image formed by pixel by pixel subtraction between image 3.2c and 3.2d (white pixel = no difference).

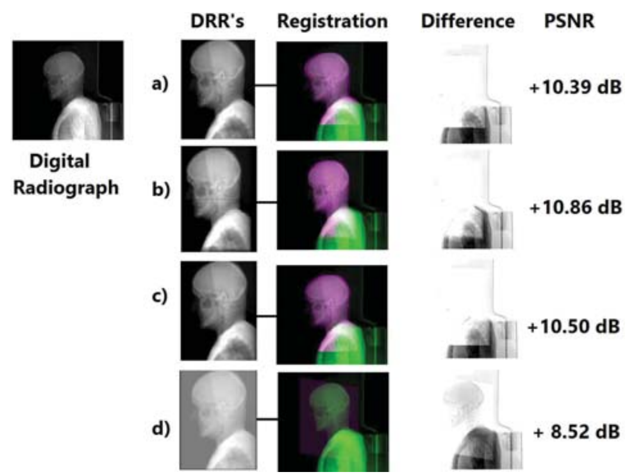

Figure 3.4. Sample *DRR's* generated using the for methods tested compared with a conventional digital radiograph generated using X-ray. The *DRR's* are computed in sagittal direction from a CT image of the head region using (a) Ray-Sum, (b) the Radon transform with cone beam, (c) the Radon transform and (d) Siddon´s Algorithm. The conventional digital radiograph is shown on the left. The next column shows *DRR's* generated from all tested methods. The next column shows difference images between the conventional digital radiograph and each method. The last column showns the PSNR values, the highest value indicates less difference between the images.

In this study, we explore the feasibility of using *DRR's* from

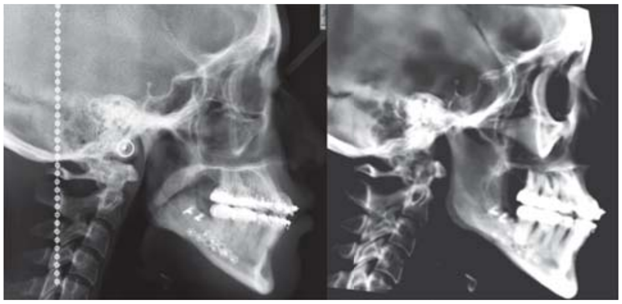

Figure 4.1. Conventional cephalogram with inherent magnification of 7.5% [6] (left) and the orthogonal cone-beam computed tomography projection without magnification (right).

CT that can be used to perform a cephalometric analysis. The *DRR* needs to be post-processed. Beside, there is a low contrast between radio-lucid and radio-opaque zones which makes landmark identification difficult and with low accuracy. The difference errors of both *DRR* were mostly by the distortion and magnification by the distances between object and source and the object and detector. The importance of generation of X-ray images from CT is to use all the information provided by only one CT scan to get explicit depth information of all structures of a patient as it is mentioned before. Advances in CBCT imaging for cephalometry will be more readily accepted by clinicians if the cephalograms can be synthesized similar to the ones they are familiar with and have used for several decades.

The orthogonal CBCT projections provided images closest to the skull and were more precise than the perspective projections by the absolute differences. Magnification and distortion in perspective projections affect the sizes of this kind of projections in comparison to the orthogonal ones and even with the real patient's head. New studies have to be developed to improve capturing protocols to get better projections for a reconstructed image. In the other hand, introducing new concepts for virtual planning and diagnosis for maxillofacial treatments, promotes the use of new technologies while leaving aside traditional methods. Virtualize orthodontics and maxillofacial diagnosis by automatic cephalometric analysis permits prior interventions and prevents improvisations while achieving more precise procedures, not only in 2D. If an acceptable cephalogram can be reconstructed, an automatic 2D and 3D cephalometric analysis can be performed and new concepts for cephalometry can be applied to three-dimensional images.

V. Conclusions

We have described four methods for generation of Digitally Reconstructed Radiographs (*DRR's*), by processing intensity-Hounsfield voxel values, in 3D CT/CBCT datasets. The slices are then projected onto a virtual detector, where the projected values are added or changed. The presented approaches do not require any pre-processing, apart from the transfer of the volume data to the *Matlab workspace* where the *DRR's* are generated. We analyzed the effects of two types of simulated x-rays beams. According to the PSNR values, both projections, orthogonal and perspective *DRR's* can be considered similar to a conventional radiograph. Therefore, they can be used to perform a cephalometric analysis. Orthogonal projection *DRR* could provide greater accuracy in the localization of mid-sagittal cephalometric landmarks than both, perspective projection *DRR* and conventional cephalometric images, because *orthogonal projection DRR's* produce matches to actual CT/CBCT that do not require any adjustment, e.g. scaling.

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Conflict of interest

The authors declare that they have no conflict of interest.

References

[1] Cobus Carstens and Neil Muller. Fast calculation of digitally reconstructed radiographs using parallelism. In *Proceedings of the Eighteenth Annual Symposium of the Pattern Recognition Association of South Africa*, pages 57–62. Durban, South Africa, 2007.

[2] Osama M Dorgham, Stephen D Laycock, and Mark H Fisher. Gpu accelerated generation of digitally reconstructed radiographs for 2-d/3-d image registration. *Biomedical Engineering, IEEE Transactions on*, 59(9):2594–2603, 2012.

[3] Rafael C Gonzalez et al. Re woods, digital image processing. *Addison–Wesely Publishing Company*, 1992.

[4] AC Kak and M Slaney. Principles of computerized tomographic imaging (piscataway, nj: Ieee). 1988.

[5] Vandana Kumar, JB Ludlow, A Mol, and L Cevidanes. Comparison of conventional and cone beam ct synthesized cephalograms. *Dentomaxillofacial Radiology*, 2014.

[6] Vandana Kumar, John Ludlow, Lucia Helena Soares Cevidanes, and André Mol. In vivo comparison of conventional and cone beam ct synthesized cephalograms. *The Angle orthodontist*, 78(5):873–879, 2008.

[7] Xiaoliang Li, Jie Yang, and Yuemin Zhu. Digitally reconstructed radiograph generation by an adaptive monte carlo method. *Physics in medicine and biology*, 51(11):2745, 2006.

[8] B De Man and S Basu. Distance-driven projection and backprojection. In *Nuclear Science Symposium Conference Record, 2002 IEEE*, volume 3, pages 1477–1480. IEEE, 2002.

[9] Mazyar Moshiri, William C Scarfe, Michael L Hilgers, James P Scheetz, Anibal M Silveira, and Allan G Farman. Accuracy of linear measurements from imaging plate and lateral cephalometric images derived from cone-beam computed tomography. *American Journal of Orthodontics and Dentofacial Orthopedics*, 132(4):550–560, 2007.

[10] P Mozzo, C Procacci, A Tacconi, P Tinazzi Martini, and IA Bergamo Andreis. A new volumetric ct machine for dental imaging based on the cone-beam technique: preliminary results. *European radiology*, 8(9):1558–1564, 1998.

[11] OsiriX. Osirix advanced open-source pacs workstation dicom viewer. http://www.osirix-viewer.com/datasets/, January 2016.

[12] Steve Pieper, Michael Halle, and Ron Kikinis. 3d slicer. In *Biomedical Imaging: Nano to Macro, 2004. IEEE International Symposium on*, pages 632–635. IEEE, 2004.

[13] Daniel Ruijters, Bart M ter Haar Romeny, and Paul Suetens. Gpu-accelerated digitally reconstructed radiographs. *BioMED*, 8:431–435, 2008.

[14] Aodong Shen and Limin Luo. Point-based digitally reconstructed radiograph. In *Pattern Recognition, 2008. ICPR 2008. 19th International Conference on*, pages 1–4. IEEE, 2008.

[15] Robert L Siddon. Fast calculation of the exact radiological path for a three-dimensional ct array. *Medical physics*, 12(2):252–255, 1985.

[16] Peter J Taub, Pravin K Patel, Steven R Buchman, and Mimis Cohen. *Ferraro's Fundamentals of Maxillofacial Surgery*. Springer, 2014.